

SMART INDIA HACKATHON 2026

Prototype

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Problem Statement Number: SIH26103

AI-Powered Integrated Project Monitoring and Risk

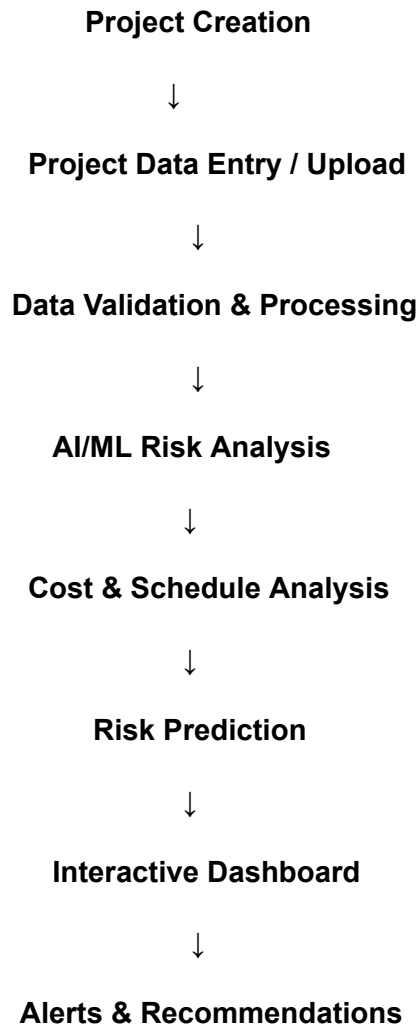
Prediction Platform

1. Prototype Objective:

The prototype demonstrates a centralized web-based platform for monitoring project progress, identifying potential risks, analysing cost and schedule deviations, and providing intelligent recommendations to project stakeholders.

The proposed system converts project-related data into easy-to-understand dashboards, risk indicators, visual analytics, alerts, and actionable insights.

2. System Workflow:



3. Prototype Screen 1 – Project Dashboard:

Dashboard:

The main dashboard provides an overview of all monitored projects.

Displayed Information:

- Total number of projects
- Projects on schedule
- Projects at risk
- Critical projects
- Overall project progress
- Cost utilization
- Schedule performance
- Recent alerts

Example:

Total Projects: 12

On Track: 7

At Risk: 3

Critical: 2

Overall Project Health: 72%

The dashboard uses charts and visual indicators to help stakeholders understand project status quickly.

4. Prototype Screen 2 – Add / Update Project:

The project manager can create a project and enter relevant project information.

Input Parameters:

- Project name
- Project category

- Project location
- Total approved budget
- Planned duration
- Start date
- Expected completion date
- Current completion percentage
- Actual expenditure
- Number of delayed activities
- Resource availability
- Milestone status

Action:

[SAVE PROJECT]

The information is stored in the project database and made available for further analysis.

5. Prototype Screen 3 – AI/ML Risk Analysis:

After project information is submitted, the system analyses the available data using machine learning and analytical techniques.

Predicted Indicators

Delay Risk: HIGH

Cost Overrun Risk: MEDIUM

Overall Project Risk: HIGH

Predicted Delay: 23 Days

Estimated Cost Overrun: ₹8.4 Lakhs

The risk values displayed in the final prototype will be generated from the trained model and project data.

6. Prototype Screen 4 – Risk Analysis & Explanation:

The system provides an explanation of the major factors contributing to project risk.

Example:

Major Risk Factors:

- ✓ Actual progress is below planned progress
- ✓ Multiple activities are delayed
- ✓ Expenditure is increasing faster than planned
- ✓ Available resources are insufficient for the current workload

Priority Risk

Schedule Delay

The system highlights the factors requiring immediate attention.

7. Prototype Screen 5 – Recommendations & Alerts:

Based on the detected risks, the platform generates actionable recommendations.

Example Alert

⚠ HIGH DELAY RISK DETECTED

The project is currently behind the planned schedule.

Suggested Actions

1. Review delayed activities.
2. Reallocate available resources.
3. Prioritize critical project milestones.
4. Review the current project schedule.
5. Monitor expenditure against remaining work.

The system can provide alerts when predefined or predicted risk levels cross critical thresholds.

8. Prototype Screen 6 – AI Project Intelligence Assistant:

The platform will include an AI-assisted interface for obtaining project insights.

Example Query

User:

"Why is Project A at high risk?"

Example Response

"Project A is currently at high risk because actual progress is below the planned schedule, several activities are delayed, and expenditure is increasing relative to the achieved progress. Immediate review of delayed activities and resource allocation is recommended."

The assistant will help stakeholders obtain project information without manually analysing multiple reports.

9. Analytics & Visualization:

The prototype will provide graphical representations of project performance.

Proposed Charts

- Planned vs. actual progress
- Planned vs. actual expenditure
- Project risk distribution
- Monthly progress
- Delay trends
- Cost escalation trends
- Project-wise risk comparison

These visualizations will help decision-makers identify deviations and trends quickly.

10. Proposed Technology Architecture:

Frontend

HTML, CSS, JavaScript

↓

Backend

Python + Flask

↓

Data Processing

Pandas + NumPy

↓

AI/ML Layer

Scikit-learn / Machine Learning Models



Database

SQLite



Visualization

[Chart.js](#)

11. Prototype Innovation:

The proposed prototype combines:

Project Monitoring + Predictive Risk Analysis + Cost & Schedule Analytics + Intelligent Recommendations

Unlike a conventional project tracking dashboard that mainly displays existing information, the proposed system aims to analyse project data and identify potential risks at an early stage.

12. Expected Prototype Output:

The prototype will demonstrate the complete flow:

Input Project Data

→ **Data Processing**

→ **AI/ML Analysis**

- **Risk Prediction**
- **Dashboard Visualization**
- **Risk Explanation**
- **Recommendations**
- **Decision Support**

Conclusion:

The proposed prototype demonstrates how an integrated AI-powered project monitoring platform can provide stakeholders with a centralized view of project performance while supporting early risk identification and informed decision-making. The prototype is designed as a scalable foundation that can be extended with real-world project datasets, advanced predictive models, automated alerts, role-based access, and additional project-intelligence capabilities.